Materials Science and Engineering Course Booklet

Bachelor's Degree Programme in Materials and Biomaterials Engineering at the University of Naples Federico II, Naples, Italy

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1

Preface

"To dress the salad, add the salt first, the lemon juice second, and the olive oil last, ***because...*** *"*

– B. Cappello, *speaking to me at the age of four*

This booklet is intended for students enrolled in the Materials Science and Engineering (MSE) course, taught in the second semester of the second year of the Bachelor's Degree Programme in Materials and Biomaterials Engineering at the University of Naples Federico II.

The booklet is designed as a guide to the course. It provides a concise presentation of the key ideas and concepts covered in the lectures. It is not intended to replace textbooks, which provide the extensive body of data, detailed explanations, examples, and exercises typically found in standard textbooks on the subject.[1].

Like introductory MSE courses taught in Materials Engineering programmes worldwide, this course adopts a science-based approach. The behaviour of materials is explained in terms of their structure, examined across

[1] A non-exhaustive list of textbooks suitable for the course is provided. Given the fundamental nature of the topics covered, earlier editions may also be used. Students are encouraged to purchase the English-language editions in order to become familiar from the outset with the technical terminology of the discipline in English.

- J.F. Shackelford, *Introduction to Materials Science for Engineers*, Pearson;
- W.D. Callister Jr., D.G. Rethwisch, *Materials Science and Engineering: An Introduction*, Wiley;
- J.C. Anderson, K.D. Leaver, R.D. Rawlings, J.M. Alexander, *Materials Science*, Chapman and Hall;
- W.F. Smith, J. Hashemi, *Foundations of Materials Science and Engineering*, McGraw Hill;
- J.-P. Mercier, G. Zambelli, W. Kurz, *Introduction to Materials Science*, Elsevier;
- M.F. Ashby, D.R.H. Jones, *Engineering Materials*, Butterworth-Heinemann;
- D. Askeland, *The Science and Engineering of Materials*, Pergamon.

different *length scales*, from the nanometre scale (10^{-9} m) to the millimetre scale (10^{-3} m). The *fil rouge* running through the course is the relationship between *structure* and *properties*. Although material properties are observed and measured at the macroscopic scale (10^{0} m), they emerge from the organization of matter across the entire hierarchy of relevant length scales. One of the most important aspects of Materials Engineering is the possibility of tailoring *some* material properties through *processing*. The operations used to transform matter and shape it into products can significantly modify its properties because they alter relevant structural features at one or more length scales. The relationships among *processing*, *structure*, and *properties* therefore provide the conceptual framework for the course (see Figure 1.1).

Although fluids play an essential role in everyday life, this course focuses on solid materials. Fluids are specifically addressed in other courses within the Bachelor's Degree Programme in Materials and Biomaterials Engineering and the Master's Degree Programme in Materials Engineering at the University of Naples Federico II.

One of the main objectives of the course is for students to become familiar with material properties. Just as most readers are familiar with the concept of *body temperature*, its units (°C), and its typical values—the physiologically relevant range being approximately 36–41 °C—Materials Science and Engineering students must become familiar with Young's modulus, yield strength, ductility, and the many other material properties introduced in the lectures. As with body temperature, familiarity with a material property requires three key pieces of information: (i) understanding the physical meaning of the property, (ii) knowing its units, and (iii) knowing its typical values (see section 1.2).

This booklet is organized into chapters that follow the sequence of topics covered in class. Depending on the topic, a chapter may correspond to one or more lectures. Some chapters include an introductory section entitled *Basic mathematical tools*, which briefly reviews the mathematical concepts needed to understand key points discussed in the chapter. Where appropriate, they also include a final section entitled *Common errors*, based on recurring student errors observed over 25 years of teaching.

1.1 Companion website

The website[2] will provide links to images, videos, news, software, and interactive tools related to MSE. Students are welcome to suggest additional resources by submitting a GitHub issue[3].

[2] MSE Booklet companion website

[3] Submit a suggestion.

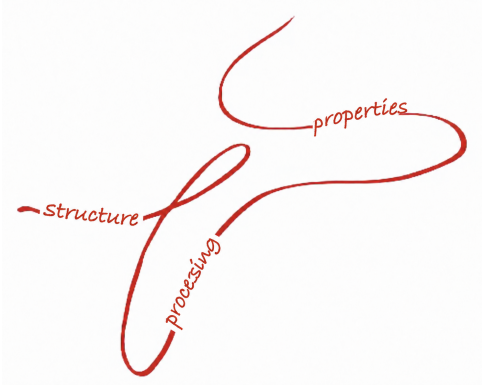


Figure 1.1: The *fil rouge* of the course: for each material property, we will identify the structural feature, at a specific length scale, that exerts the dominant influence on that property. When modification of that structural feature is possible, processing may be used to alter it and, consequently, the associated property.

1.2 Common errors

Statements such as "My body temperature is green", "My body temperature is 36 km/h" or "My body temperature is 36,000 °C" would immediately reveal a lack of familiarity with the property. The three statements respectively indicate a failure to understand its physical meaning, its units, and its typical values: in each case, one of the three key pieces of information is missing.

2

Which materials are available?

2.1 Many requirements

Why do material properties matter? Because the performance of the components and products we use in everyday life depends on the properties of the materials from which they are made. Throughout the course, we will examine these relationships, which are not always straightforward.

How many material properties do we need to study? Unfortunately, quite a few. A component may be required to meet numerous structural and functional requirements, where *functional* refers to all non-structural requirements. Failure to satisfy even one of them will compromise the entire design. This might suggest that all material properties—and all requirements—are equally important in a given component. However, every design generally has a single *limiting property*, that is the material property that ultimately prevents further improvement of the design with respect to a chosen objective. To understand this concept, we must first clarify what is meant by "improving a design".

Standards and regulations establish requirements for the safety, reliability, and acceptable behaviour of the products that surround us. Although these rules may appear restrictive, they represent one of the major achievements of the modern world. A chair, for example, must satisfy all applicable requirements: among other things, it must support a specified load without failure or excessive deformation. Once these requirements have been met, improving the design does **not** mean making the chair stronger or stiffer. Performance beyond the required level may provide no practical benefit. Instead, improvement may mean reducing mass, cost, or environmental impact while continuing to satisfy every requirement. Suppose, for example, that the objective is to reduce the mass of a chair. As the design becomes lighter, a point will eventually be reached at which one of the requirements

is no longer satisfied. If the chair deforms excessively, stiffness is the limiting design constraint, and the corresponding material property—such as Young's modulus—becomes the limiting factor in materials selection.

Identifying the limiting property of a specific design is not straightforward: it requires detailed knowledge of the product and considerable engineering experience. Nevertheless, it is important because it (i) guides materials selection and (ii) identifies the most efficient directions for materials research and development aimed at innovation in that field, which is critically important for materials scientists and engineers.

2.2 Many materials

Today, engineers can choose from well over 100,000 materials. This enormous number may seem daunting. How can we navigate such a vast range of possibilities, and how can we select the most suitable material for a given application?

Fortunately, materials can be organized into classes and subclasses, allowing us to approach selection through a hierarchical decision process. The four main classes are *metals*, *ceramics*, *polymers*, and *composites*. For the purposes of this course, we will also consider *semiconductors* as a separate class because of the importance of their electrical behaviour. We will therefore refer to five groups: four broad classes traditionally associated with structural materials, together with semiconductors.

This classification also reflects the historical emphasis placed on structural behaviour. Functional materials do not form an equivalent set of separate classes: electrical, magnetic, optical, and other functional behaviours may be exhibited by materials belonging to several of the classes listed above.

Engineers use several tools to support the selection of materials for specific applications. Databases, charts, and tables of many different types are available (see the MSE Booklet companion website).

Materials scientists and engineers make extensive use of the so-called *Ashby charts*. These charts plot one material property against another, while the different classes of materials occupy characteristic regions of the diagram, often referred to as *material-class envelopes*. An example is shown in Figure 2.1. Ashby charts provide an immediate visual map of the materials world and facilitate comparisons among different classes. As we will see in the sections on *materials selection*, when combined with the requirements and objectives of a design, they help identify the materials best suited to a specific application.

Ashby charts generally use logarithmic scales because material properties typically span several orders of magnitude. On linear scales, classes characterized by comparatively low values are compressed near the origin,

making their ranges and differences difficult to distinguish. To illustrate this effect, Figure 2.2 presents the same data and material-class envelopes as Figure 2.1, but on linear scales. Only the regions corresponding to ceramics and metals, which exhibit comparatively high densities and Young's moduli, remain readily distinguishable.

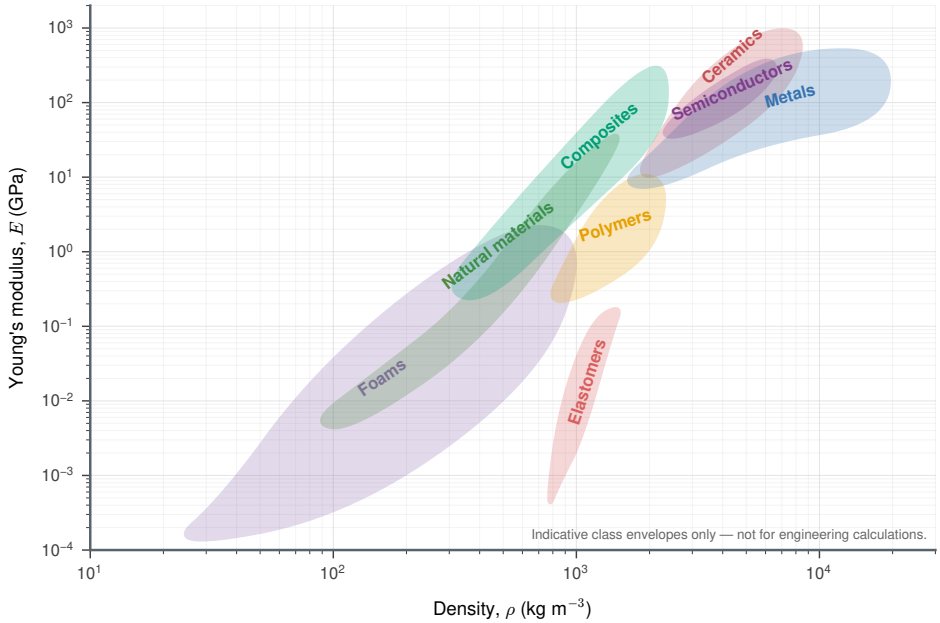


Figure 2.1: Schematic Ashby chart of Young's modulus versus density.

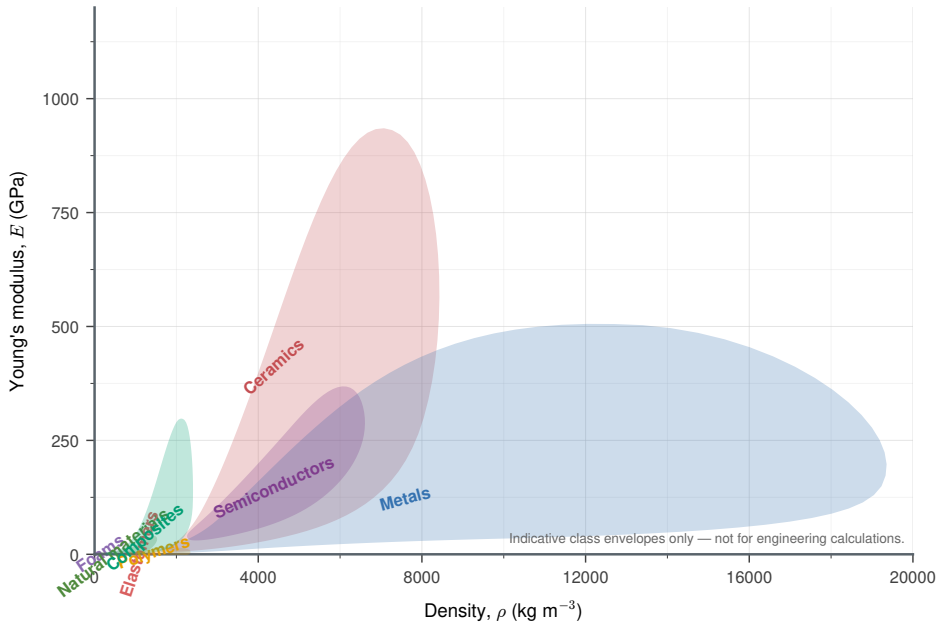


Figure 2.2: Schematic Young's modulus versus density on linear scales. Most material classes are compressed near the origin because the properties span several orders of magnitude, making comparison and materials selection difficult.

3

Atomic bonding

3.1 Basic mathematical tools

We will examine how the force between two ions varies with their separation and how this relationship determines their equilibrium distance. Familiarity with functions is therefore useful. In this context, the usual notation $y = f(x)$ is replaced by $F = f(a)$, where F, measured in Newtons, N, is the force that the two ions exert on each other and a, measured in metres, m, is the distance between them. Nevertheless, these are the same types of functions studied in *Calculus*. Let us first recall the shapes of the functions $y = \frac{1}{x^2}$ and $y = \frac{1}{x^7}$, shown in Figure 3.1.

If we restrict the vertical axis to the range $(0 < y < 10)$ and replace $y = \frac{1}{x^7}$ with its opposite, we obtain the graph shown in Figure 3.2.

Adding the functions $y = -\frac{1}{x^7}$ and $y = \frac{1}{x^2}$ curves, or equivalently subtracting $y = \frac{1}{x^7}$ from $y = \frac{1}{x^2}$ curves), gives $y = \frac{1}{x^2} - \frac{1}{x^7}$, as shown in Figure 3.3.

Finally, let us recall the concept of an antiderivative, also called a primitive function. Figure 3.4 shows the function $y = \frac{1}{x^2} - \frac{1}{x^7}$ and one of its antiderivatives, obtained by setting the integration constant $C = 0$. The same curves are shown on different scales in Figure 3.5, making the minimum of the antiderivative more clearly visible. Equivalently, $y = -\frac{1}{x} + \frac{1}{6x^6}$ can be regarded as the derivative of $y = \frac{1}{x^2} - \frac{1}{x^7}$ as its derivative. The derivative is equal to zero at $x = 1$, where the antiderivative has a local minimum.

3.2 Ionic bonding, forces and energies

Two oppositely charged ions (e.g., (Na^+) and (Cl^-) attract each other through the Coulomb force, whose magnitude is proportional to the inverse square of the distance between them. A graph of the attractive force (F_A) as a function of the interionic distance (a) therefore has the same shape as

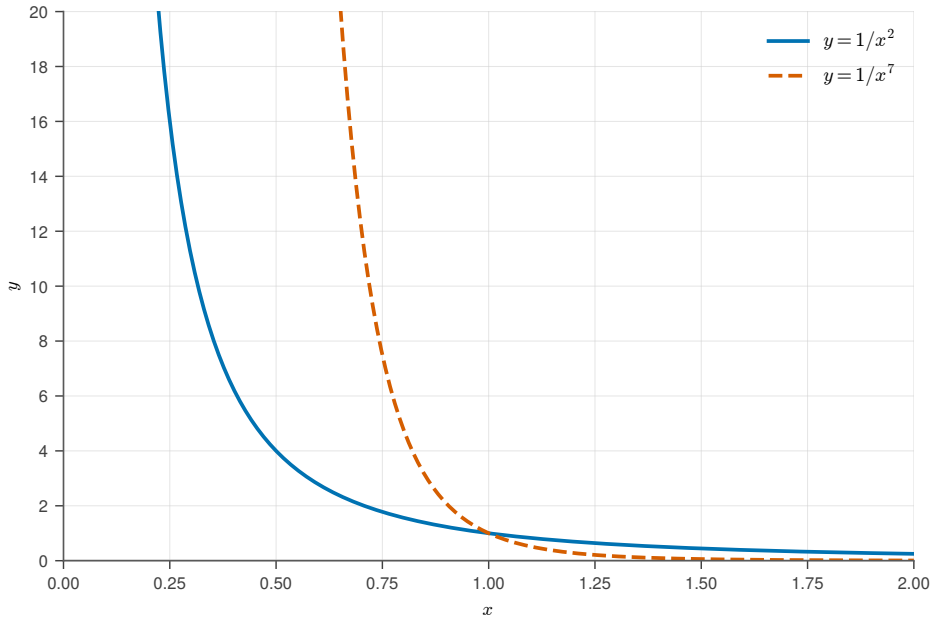


Figure 3.1: Graphs of the functions $y = \frac{1}{x^2}$ in blue and $y = \frac{1}{x^7}$ in orange.

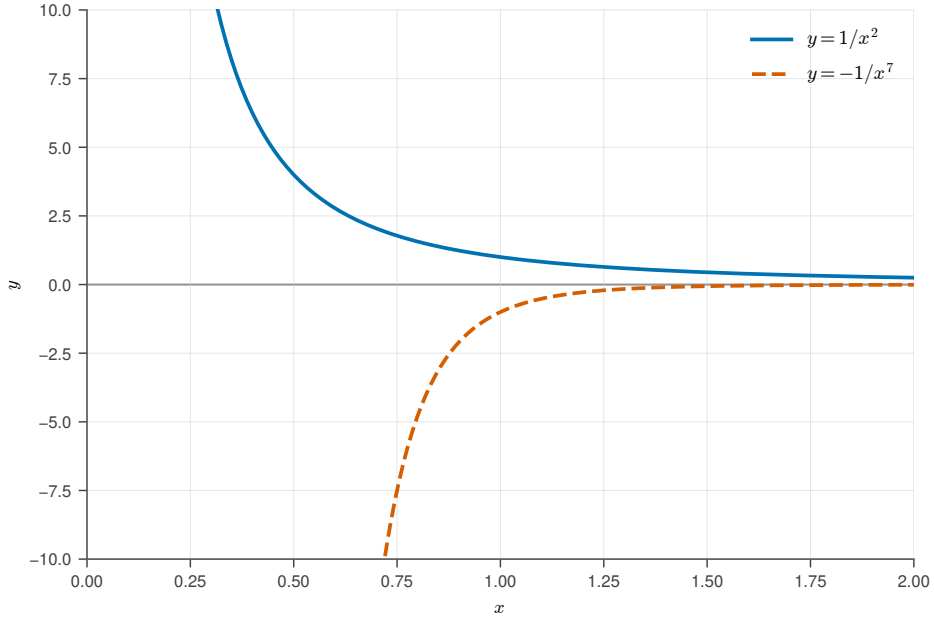


Figure 3.2: Graphs of the functions $y = \frac{1}{x^2}$ in blue and $y = -\frac{1}{x^7}$ in orange.

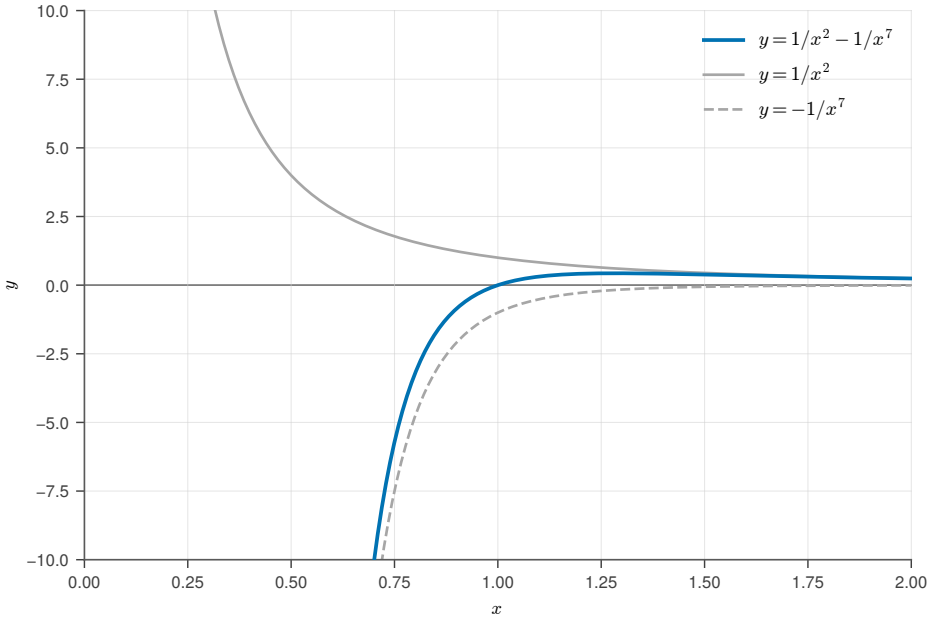


Figure 3.3: Graph of the function $y = \frac{1}{x^2} - \frac{1}{x^7}$, together with its component functions.

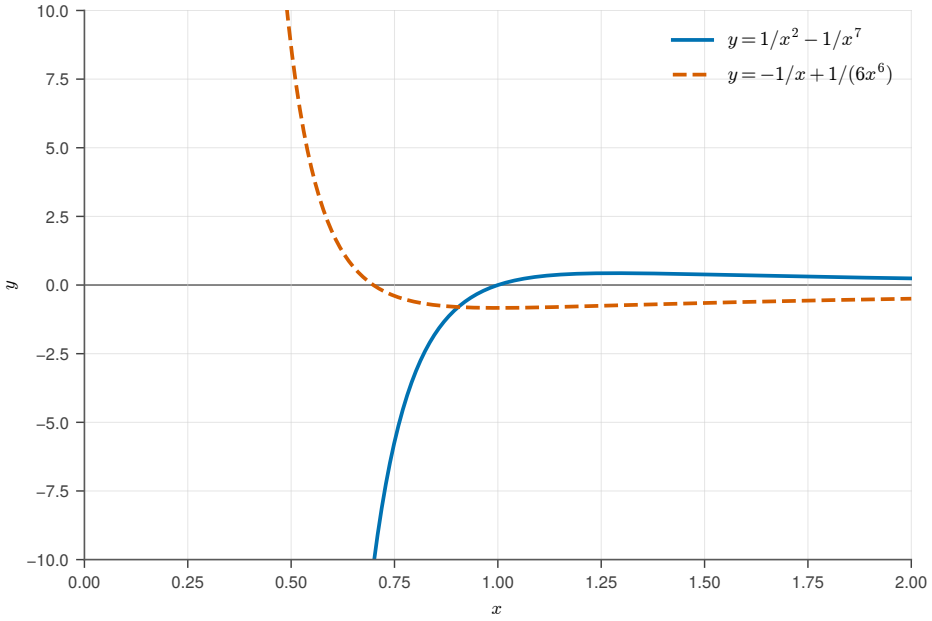


Figure 3.4: Graphs of the functions $y = \frac{1}{x^2} - \frac{1}{x^7}$ in blue and $y = \int \left(\frac{1}{x^2} - \frac{1}{x^7}\right) dx = -\frac{1}{x} + \frac{1}{6x^6} + C$, with $C = 0$ in orange.

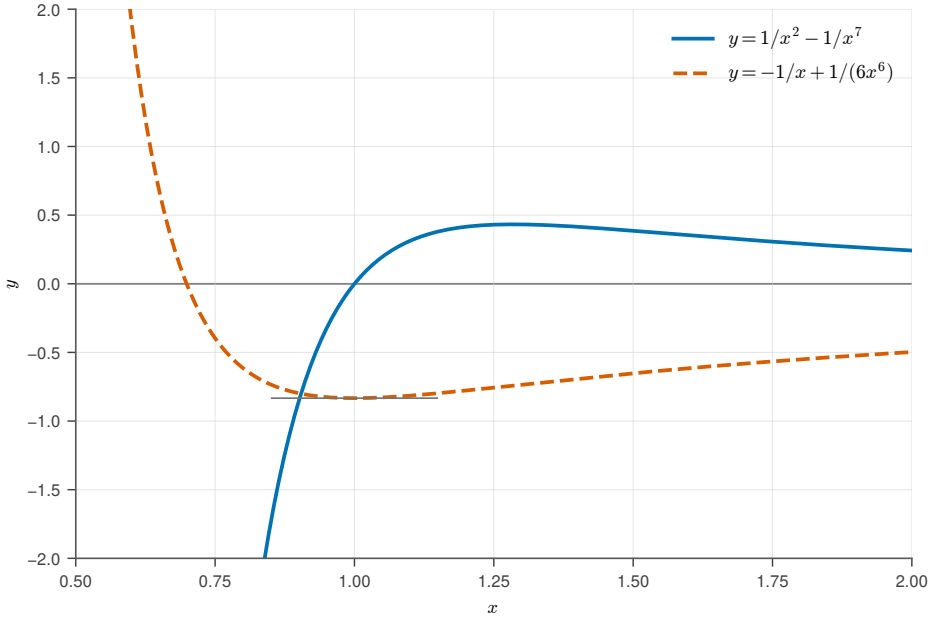


Figure 3.5: Graph of the functions $y = \frac{1}{x^2} - \frac{1}{x^7}$ in blue and $y = \int \left(\frac{1}{x^2} - \frac{1}{x^7}\right) dx = -\frac{1}{x} + \frac{1}{6x^6} + C$, with $C = 0$ in orange. Different axis scales from those used in the previous figure make the minimum of the orange curve and the corresponding zero of the blue curve at $x = 1$ more clearly visible.

the $1/x^2$ curve in Figure 3.1, with a on the horizontal axis and F on the vertical axis.[1]

Because the attractive force increases as the ions approach each other, one might expect them eventually to collide. At very short distances, however, their electron clouds begin to overlap, giving rise to a repulsive force (F_R) that increases very rapidly as the interionic distance decreases. In the present simplified treatment, this short-range repulsion is modelled by a steep inverse-power function.[2] The terms "short-range" and "steep" indicate that, at sufficiently short distances, the repulsive force increases abruptly, as though the counterion had encountered a wall.

The $1/x^7$ function (the orange curve in Figure 3.1) is very small—almost negligible—at large x, where it lies well below the $1/x^2$ curve. As x decreases, however, it increases much more rapidly and eventually exceeds the $1/x^2$ curve. At $x = 1$, the two functions coincide and both have the value 1. If coefficients other than unity multiply $1/x^2$ and $1/x^7$, the intersection no longer necessarily occurs at $x = 1$, and the values of the two functions change, but their characteristic shapes remain the same.

We can therefore state that the attractive force prevails at large distances, whereas the repulsive force prevails at short distances. Equilibrium is reached where the two forces have equal magnitudes and opposite directions, so that their sum is zero. This is more clearly illustrated in Figure 3.3. The light-blue curve represents the sum of $1/x^2$ and $-1/x^7$, and therefore the balance between the two forces, using the convention that the attractive force is positive and the repulsive force is negative. In this dimensionless example, equilibrium occurs at $x = 1$, where the two contributions have the values $+1$ and -1, respectively. For real ions, the force terms have analogous dependences on distance, and their sum becomes zero at a specific interionic distance. Figure 3.6 shows a representative dimensional model for (Na^+) and (Cl^-), with force expressed in newtons and interionic distance in metres.

This force-curve representation also helps explain the stability of the equilibrium. Suppose that an external force F' displaces the ions from their equilibrium separation a_0 to a new distance a', for example by pulling them farther apart. At this distance, the interionic restoring force has the same magnitude F' but acts in the opposite direction. If the external force is removed, the restoring force drives the ions back towards their equilibrium separation. Conversely, if the ions are pushed closer together, the system moves onto the negative branch of the curve, where the repulsive force drives them apart. Thus, on either side of the equilibrium position, the net interionic force acts as a restoring force (see Figure 3.7). This provides a first

[1] This situation can be visualized by placing one ion at the origin and allowing the counterion to move along the horizontal axis.

[2] A more accurate description of short-range repulsion is commonly provided by an exponential function.

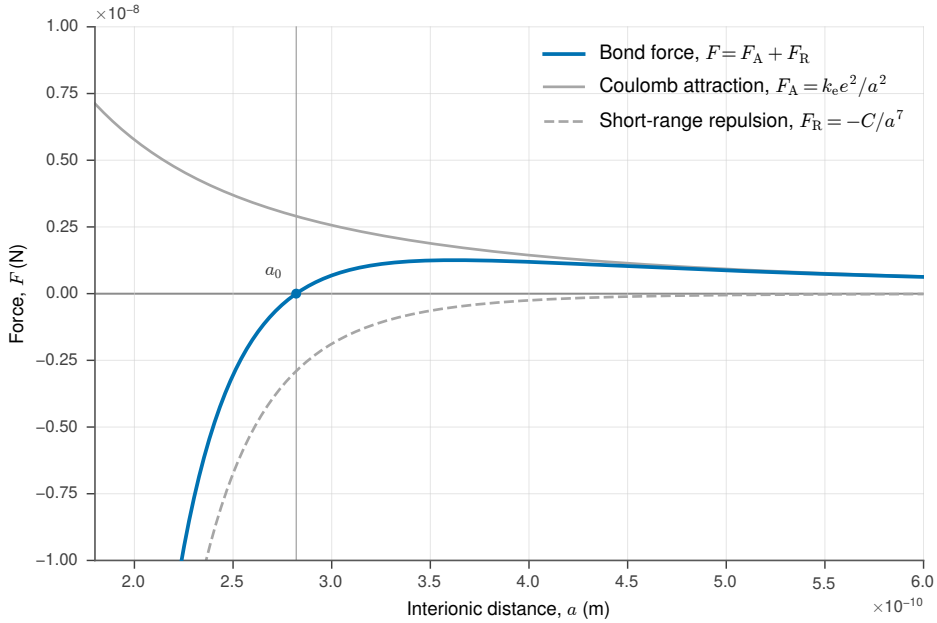


Figure 3.6: Graph of the repulsive and attractive interactions between (Na^+) and (Cl^-) ions as well as their resultant force as a function of their separation distance. a_0 is the equilibrium distance between the two ions.

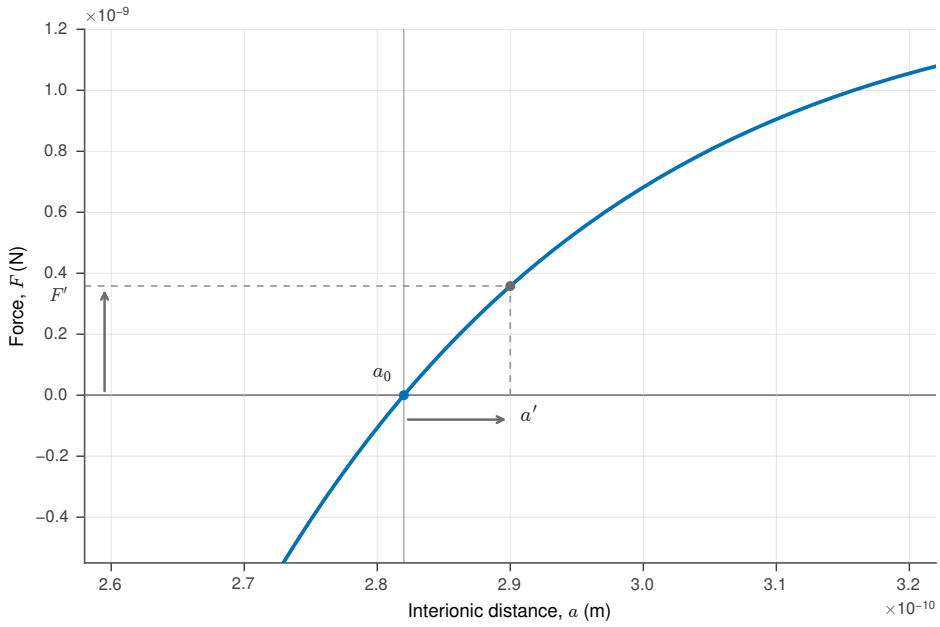


Figure 3.7: Effect of a tensile force F' on the distance between two ions increasing from the equilibrium a_0 to a'

example of how atomic-scale bonding behaviour—in this case, the slope of the interionic force–distance curve near the equilibrium distance—is related to a material property observed at the macroscopic scale: the elastic modulus. Throughout this booklet, such connections will be highlighted as *structure to property* relationships, which are central to the study of MSE.

Consider, for example, a tensile force applied to a piece of solid NaCl. The resulting elongation of the specimen is accompanied, at the atomic scale, by small changes in the distances between many neighbouring ions. Each displaced ion pair develops a restoring force of the type described above. Because a macroscopic specimen contains an order of Avogadro's number of interacting ion pairs, their collective response produces a macroscopic restoring force. Provided that the deformation remains within the elastic regime, the specimen returns to its original dimensions when the external force is removed. This is the basic idea of elasticity, which will be defined more rigorously in the Chapter 8.

Another fundamental point is that the bonding forces between the ions are intrinsic to a given material and cannot be changed. Consequently, its elastic moduli—as we will see later, there is more than one—are also intrinsic properties and cannot be modified.

Now, let us consider another ionic solid characterized by different interionic forces, represented by the red curve in Figure 3.8. The same applied force F' produces a larger ionic displacement for the red curve than for the blue curve. We will see that this behaviour can be described in terms of the red material having a lower elastic modulus than the blue material. Equivalently, the same comparison can be made using a *dual* approach: if the same displacement is imposed on both materials, the restoring force is greater for the blue material (see Figure 3.9).

Finally, by recalling the relationship between bonding force and bonding energy, we can describe the same phenomena using the bonding-energy curve as a function of interionic distance (see Figure 3.5). The slope of the force curve at the equilibrium distance is related to the curvature of the energy curve at its minimum: the greater the curvature, the stiffer the bond. The depth of the energy well represents the energy required to separate the bonded ions (see Figure 3.10). In an NaCl crystal, melting involves overcoming the interactions that maintain the ordered ionic structure. Stronger ionic bonding therefore generally requires more thermal energy to melt the material and is associated with a higher melting temperature. This provides another *structure to property* example: the relationship between ionic bonding and melting temperature. In the chapter on the thermal properties of materials, we will also see how the coefficient of thermal expansion is related to the asymmetry of the bonding-energy curve. It follows that both melting temperature and the coefficient of thermal expansion are intrinsic properties of a given material and cannot be modified.

We have explored the relationship between atomic bonding and mate-

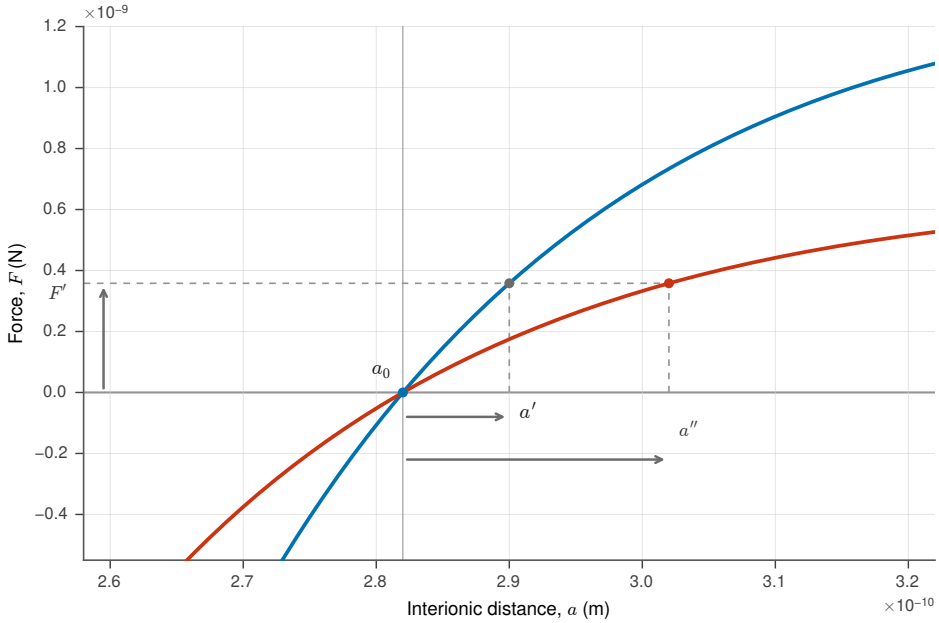


Figure 3.8: Effect of the same tensile force F' on the interionic distance in two different materials. The steeper blue curve represents stronger bonding and therefore a smaller displacement, from a_0 to a', than the red curve, for which the distance increases from a_0 to a''.

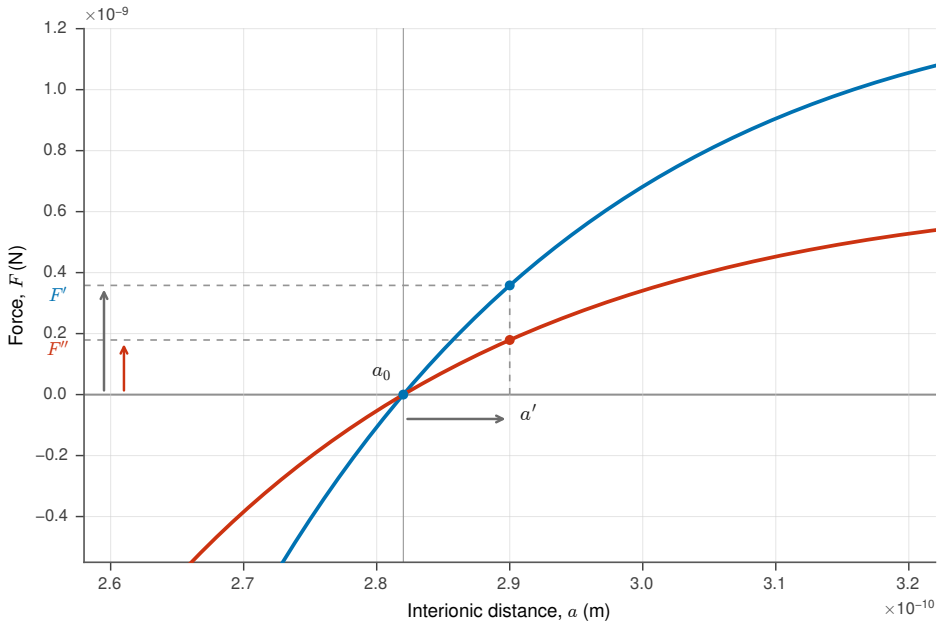


Figure 3.9: Effect of the same displacement, from a_0 to a', on the interionic force in two different materials. The steeper blue curve represents stronger bonding and therefore requires a larger force F' to produce this displacement, whereas the material represented by the red curve requires the smaller force F''.

rial properties using ionic bonds and ionic solids as our primary examples. However, the same general considerations also apply to covalent, metallic, and secondary bonds. Readers are encouraged to consult the corresponding chapters of the recommended textbooks, familiarise themselves with the relevant data, and work through the problems provided therein.

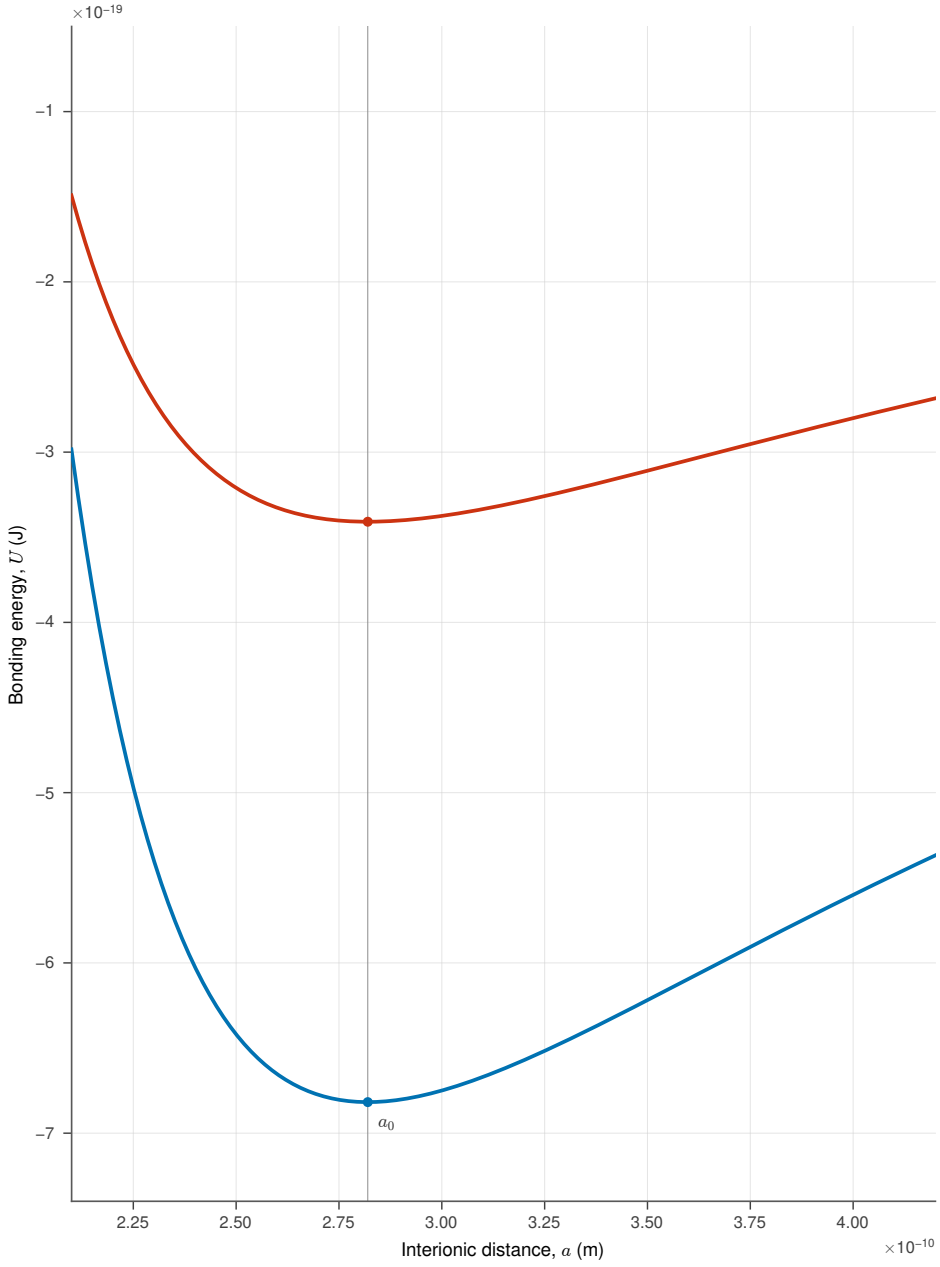


Figure 3.10: Bonding-energy curves for two materials with different elastic moduli, corresponding to those shown in Figure 3.8

4

Crystal structure

4.1 Basic mathematical tools

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Crystal defects

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Phase diagrams

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Thermal treatments

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Mechanical behaviour

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Thermal properties

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Electrical properties

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Materials Sustainability

Glossary

coefficient of thermal expansion measure of the dilation effect induced by temperature increase. 15

elastic modulus measure of a material resistance to elastic deformation. 15

materials selection examples of how to select the best material in a specific application. 6

structure to property examples of how details at a specific scale of the material structure affect a specific property. 15

Acronyms

MSE Materials Science and Engineering. 1, 2, 15